



FitPulse: Health Anomaly Detection using Fitness Devices

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Project Overview & Problem Statement

What Fitness Devices Track

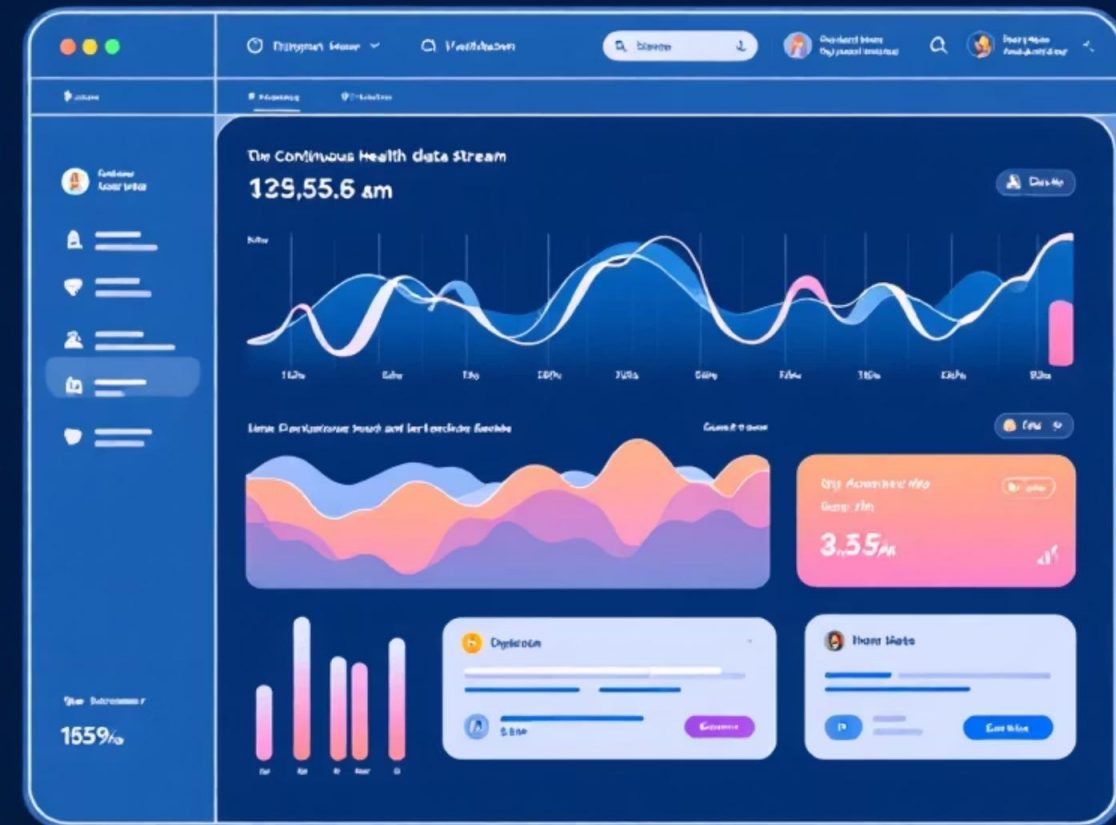
Modern wearable devices continuously monitor multiple health metrics throughout the day:

- Heart rate (beats per minute)
- Physical activity (steps, distance)
- Calorie expenditure
- Sleep patterns and quality

The Challenge

Fitness devices generate massive amounts of continuous health data, creating an overwhelming volume of information. Manual monitoring of this data stream is practically impossible for individuals and healthcare providers.

Our solution: FitPulse provides an automated, intelligent system that transforms overwhelming fitness tracker data into actionable health insights.



Project Objectives

FitPulse aims to bridge the gap between continuous health data collection and actionable health insights through automated anomaly detection. Our system is designed to identify irregular patterns that may indicate potential health concerns, enabling early intervention and preventive care.

01

Detect Abnormal Health Patterns

Identify deviations from normal health metrics using fitness device data

03

Multi-Type Anomaly Identification

Recognize point anomalies (sudden spikes), contextual anomalies (unusual patterns), and collective anomalies (sustained irregular behavior)

02

Efficient Time-Series Processing

Handle and analyze sequential health data with temporal dependencies

04

Interactive Visual Insights

Provide clear, accessible visualizations and an interactive dashboard for exploring detected anomalies

Data Description & Structure

Data Sources

Our system processes health data collected from fitness wearables and smartwatches. The data is structured in standard formats (CSV/JSON) to ensure compatibility across different device manufacturers and easy integration into the analysis pipeline.

The time-series nature of this data captures the temporal evolution of health metrics, allowing us to understand patterns, trends, and deviations over time.

Our dataset captures comprehensive health metrics every 5 minutes over a 7-day period, providing high-resolution insights into daily patterns and physiological responses.



Timestamp

Precise temporal markers for each measurement



Heart Rate (BPM)

Beats per minute captured continuously throughout the day



Steps & Activity

Physical activity levels and movement patterns



Sleep Duration

Sleep quality and duration metrics for recovery analysis

📌 **Key Characteristic:** All features are organized as time-series data, maintaining temporal order and enabling sophisticated pattern recognition across different time scales.

Key points about dataset

- **Data Source:** Smartwatch / Fitness tracker
- **Data Format:** CSV / JSON
- **Health Metrics Collected:**
 - ❖ Heart Rate (BPM)
 - ❖ Sleep Duration (hours)
 - ❖ Step Count
 - ❖ Calories Burned
 - ❖ Activity Type
- **Timestamp Resolution:** 5-minute intervals
- **Duration:** 7 days (~2,016 records per metric)

High-resolution data helps capture daily health patterns accurately.



Tools & Technologies Used

Programming & Libraries:

Python – core programming language

Pandas, NumPy – data handling and preprocessing

Matplotlib, Plotly – data visualization

Scikit-learn – clustering and machine learning models

Time-Series & Modeling

TSFresh – feature extraction from time-series data

Facebook Prophet – trend and seasonality modeling

KMeans & DBSCAN – pattern grouping and outlier detection

Deployment & Data

Streamlit – interactive web dashboard

Data Formats – CSV / JSON from fitness trackers

Milestone 1: Data Collection & Preprocessing

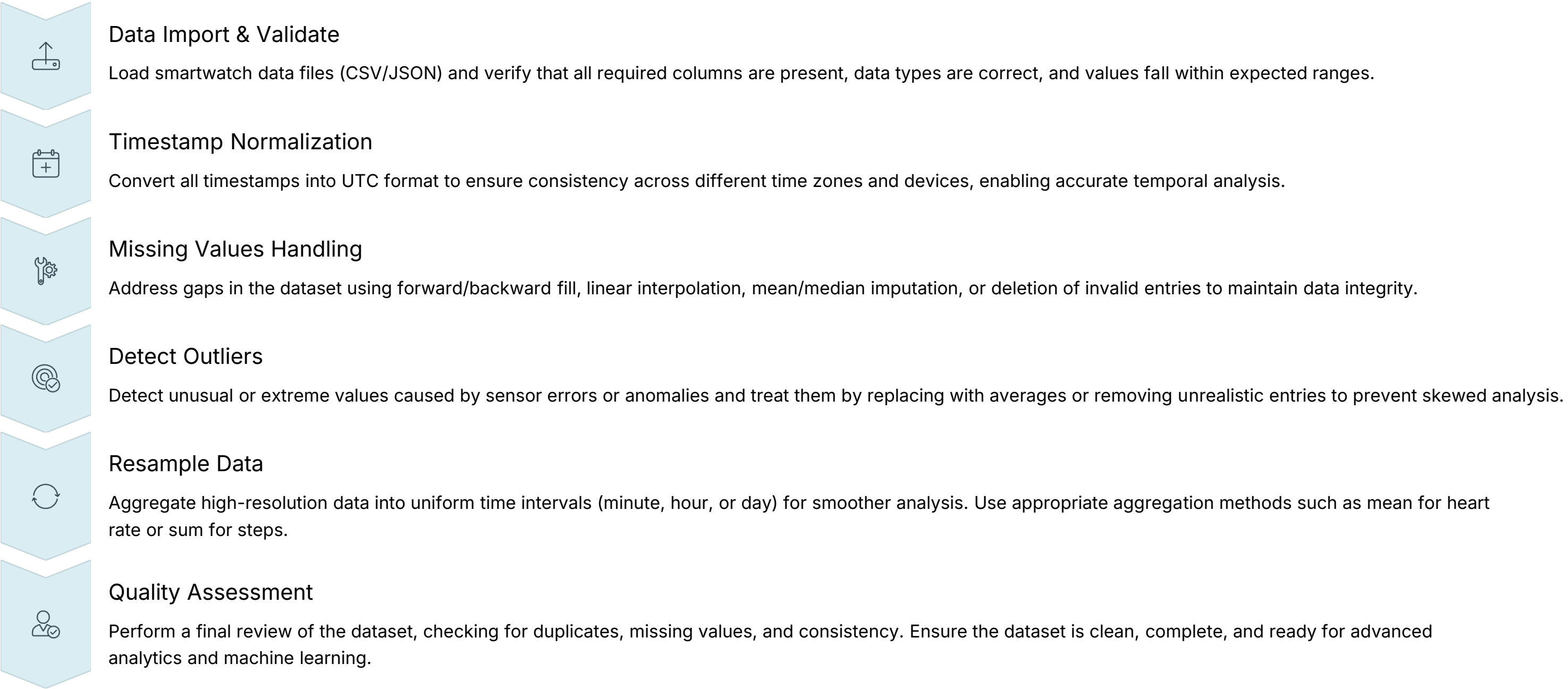
- Collect raw smartwatch data from fitness devices
- Preprocess wearable data to create a **clean, structured time-series dataset**
- Prepare data for:
 - ❖ Pattern analysis
 - ❖ Anomaly detection
 - ❖ Interactive dashboards

Final output:

- Clean and consistent time-series dataset
- Uniform time intervals
- Ready for further milestones.



Preprocessing Pipeline



Milestone 2: Feature Extraction & Modeling

- Analyze preprocessed time-series fitness data
- Convert raw sensor signals into **meaningful statistical features**
- Learn **normal behavior patterns** using forecasting & clustering
- Prepare analytical foundation for **anomaly detection**

Key Techniques Used

- TSFresh for feature extraction
- Prophet & Rolling Median for trend modelling
- KMeans & DBSCAN for behavioral clustering



Feature Extraction using TSFresh

TSFresh (Time Series Feature extraction based on scalable hypothesis tests) automatically extracts hundreds of statistical features from time-series windows, enabling machine learning on temporal data.

01

Window Creation

Created sliding windows of 12 data points, representing 1 hour of continuous monitoring

02

Feature Extraction

Applied MinimalFCParameters to extract mean, median, standard deviation, entropy, and autocorrelation

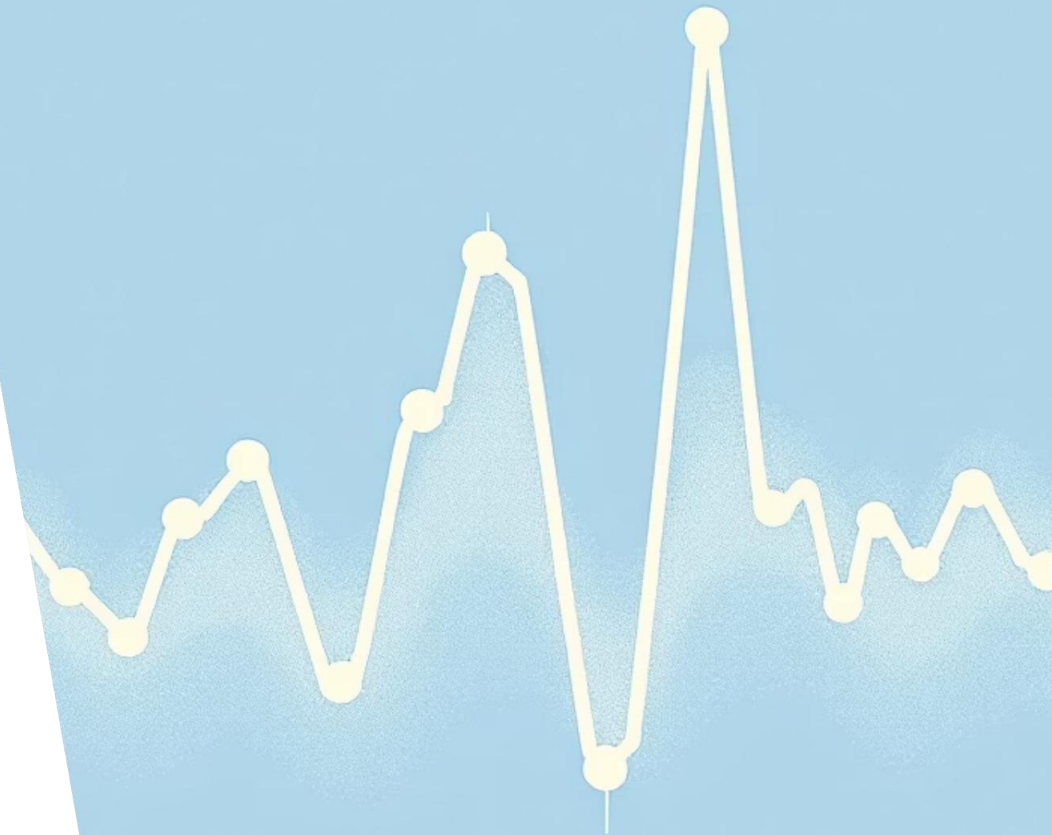
03

Feature Selection

Removed constant and non-informative features to reduce dimensionality



Output: A clean feature matrix ready for clustering and anomaly detection modeling



Trend Modeling Approaches

Identifying baseline patterns is essential for detecting deviations that may indicate health anomalies.

Primary: Facebook Prophet

Advanced forecasting tool designed for time-series with strong seasonal patterns and multiple seasonality

- Captures daily and weekly trends
- Handles missing data gracefully
- Provides uncertainty intervals

Backup: Rolling Median

When Prophet encountered environment constraints, rolling median provided a robust alternative

- Smooths noisy sensor data
- Captures typical behavior patterns
- Enables 3-sigma anomaly detection



Clustering for Behavioral Patterns

Clustering algorithms group similar time windows together, revealing distinct behavioral patterns in fitness data without labeled training data.



KMeans Clustering

Partitions data into fixed number of clusters, ideal for identifying distinct activity levels

- Groups windows into 3 behavioral clusters
- Uses PCA for dimensionality reduction
- Visualizes patterns in 2D space



DBSCAN Clustering

Density-based approach that identifies outliers and noise points automatically

- No predefined cluster count required
- Detects noise and outlier windows
- Robust to irregular cluster shapes

Milestone 2 Outcome

- Successfully converted raw time-series fitness data into **meaningful statistical features** using TSFresh
- Learned **normal behavior patterns** in heart rate data through:
 - ❖ Time-series forecasting (Prophet / Rolling Median)
 - ❖ Unsupervised clustering (KMeans & DBSCAN)
- Identified deviations from normal patterns using **statistical and model-based approaches**

Milestone 2 transforms clean health data into intelligence by extracting features and learning normal behavior patterns, which are essential for accurate anomaly detection.



Milestone 3: Anomaly Detection & Visualization

Objective:

To identify abnormal health patterns in fitness data by leveraging learned behavior models and extracted features.

Key points:

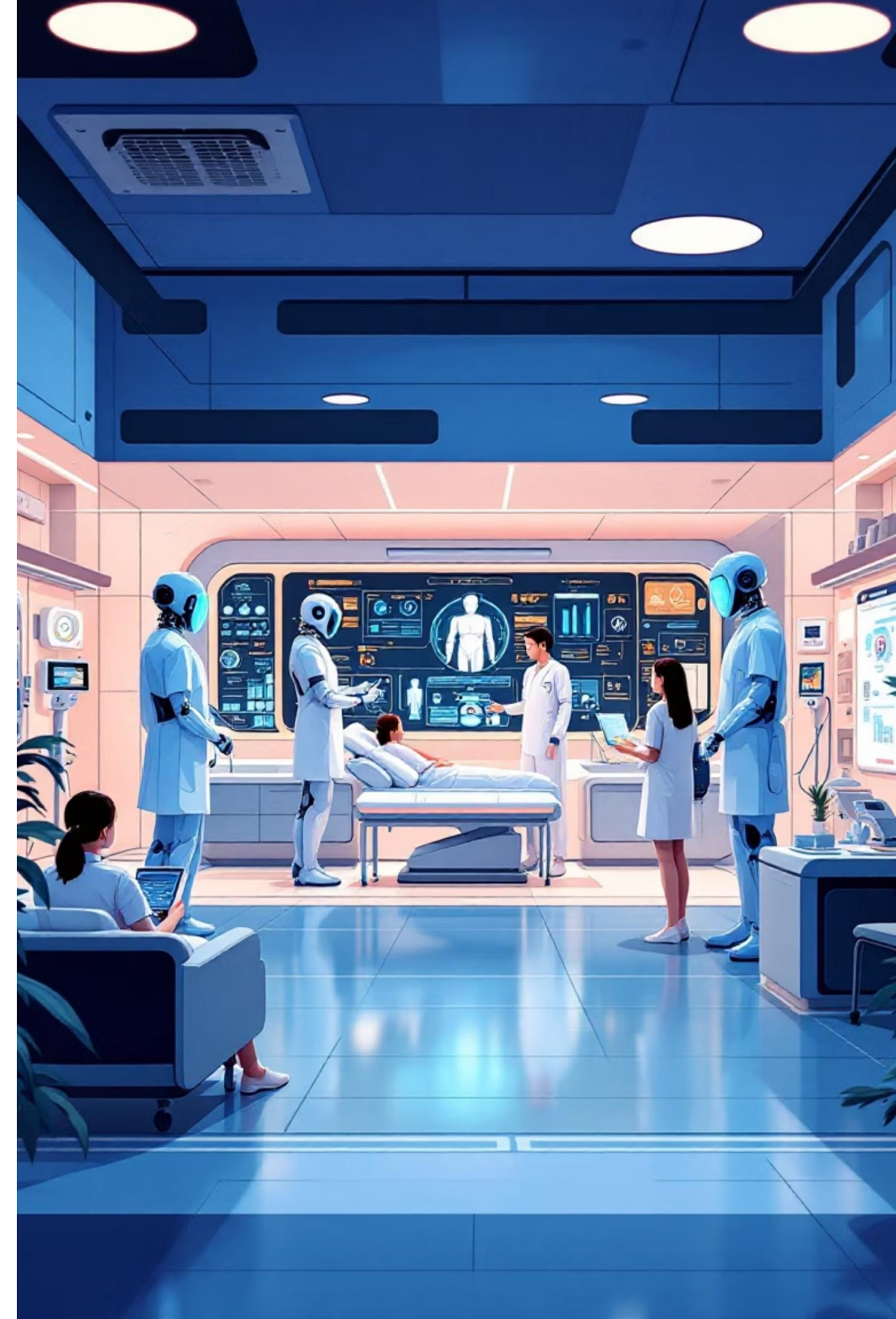
- Applies insights generated in **Milestone 2**.
- Detects unusual deviations in heart rate, activity, and sleep data
- Converts analytical results into meaningful anomaly indicators
- Prepares anomaly outputs for visualization and dashboard integration

Role:

- Transforms patterns into actionable health alerts
- Enables early identification of potential health risks

Outcomes of Milestone 3:

- Successfully detected anomalies in fitness time-series data
- Identified abnormal patterns in heart rate, activity, and sleep
- Visualized anomalies clearly using time-series plots
- Generated interpretable insights ready for dashboard integration



Understanding Health Anomalies

An **anomaly** is a data point or pattern that deviates significantly from expected behavior, potentially indicating health concerns requiring attention. Different anomaly types require **different detection methods**.

Point Anomaly

A single data point that is unusual compared to the rest of the dataset.

Example: Heart rate suddenly spikes to 180 bpm during rest.

Contextual Anomaly

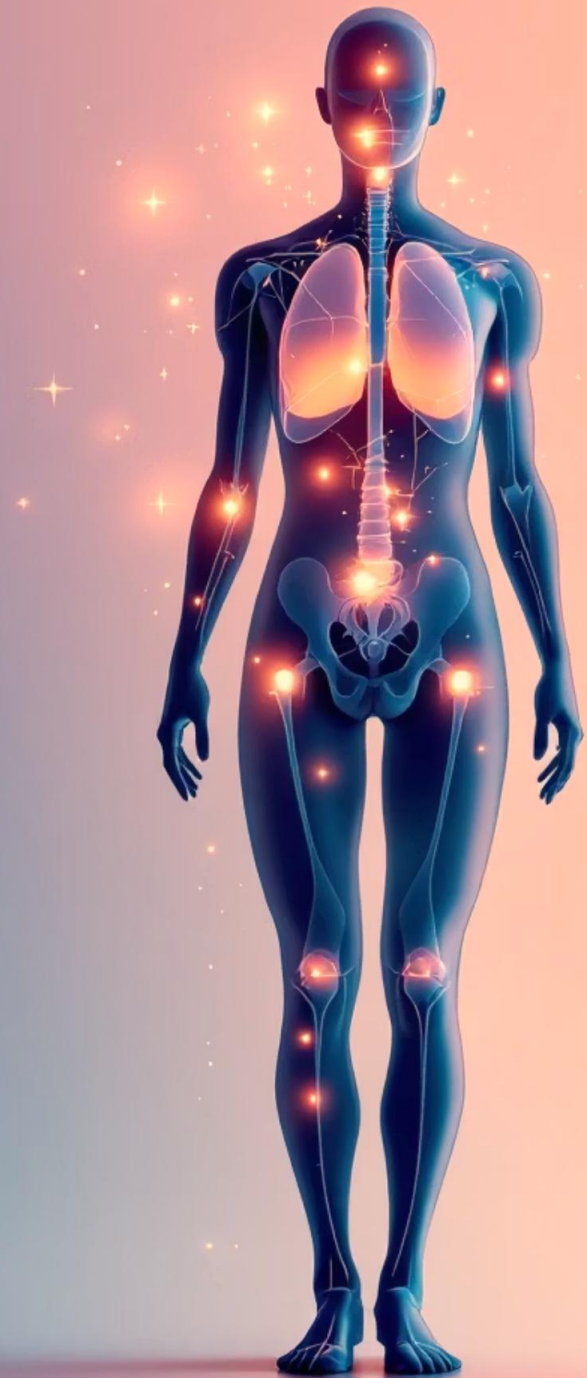
A data point that is anomalous only in a specific context or time period.

Example: High heart rate during typical exercise hours.

Collective Anomaly

A collection of data points that together form an unusual pattern.

Example: Elevated heart rate with zero movement for extended periods.



Anomaly Detection Approach

No single anomaly detection method can capture all types of health irregularities. Our solution combines three complementary techniques to provide comprehensive coverage.

Rule-Based Detection

1

Uses predefined health thresholds (e.g., heart rate > 180 BPM or < 40 BPM) to flag immediate concerns. Simple but effective for extreme values. Detects **Point Anomalies**

2

Model-Based Detection

Compares actual values against Prophet forecasts. Large residuals (differences between predicted and actual) signal unexpected deviations from normal patterns. Detects **Contextual Anomalies**

Clustering-Based Detection

3

Identifies points that fall outside normal behavioral clusters. DBSCAN marks isolated points as noise, while KMeans flags points far from cluster centroids. Detects **Collective / Behavioral Anomalies**

Rule-Based Anomaly Detection

Detecting Point Anomalies Through Threshold Logic

Rule-based detection applies predefined medical thresholds to identify extreme physiological values quickly and interpretably. This method excels at catching obvious health concerns in real-time.

What it does:

1. Uses **medical and logical thresholds**
2. Flags extreme values immediately

Detection Rules Implemented

- **High heart rate:** Greater than 100 bpm at rest
- **Low heart rate:** Below 40 bpm during waking hours
- **Minimal activity:** Fewer than 100 steps during active periods
- **Extreme calorie burn:** Values exceeding 99th percentile

Key Advantage:

Good for catching sensor errors and extreme risks, Fast, interpretable



Model-Based Anomaly Detection

Why rules are not enough

- Different users have different “normal” heart rates
- Same value can be normal for one person, abnormal for another

1

Learn Normal Patterns

Prophet analyzes historical data to understand typical daily, weekly, and seasonal health rhythms.

2

Generate Predictions

The model forecasts expected values based on learned patterns and current context.

3

Calculate Residuals

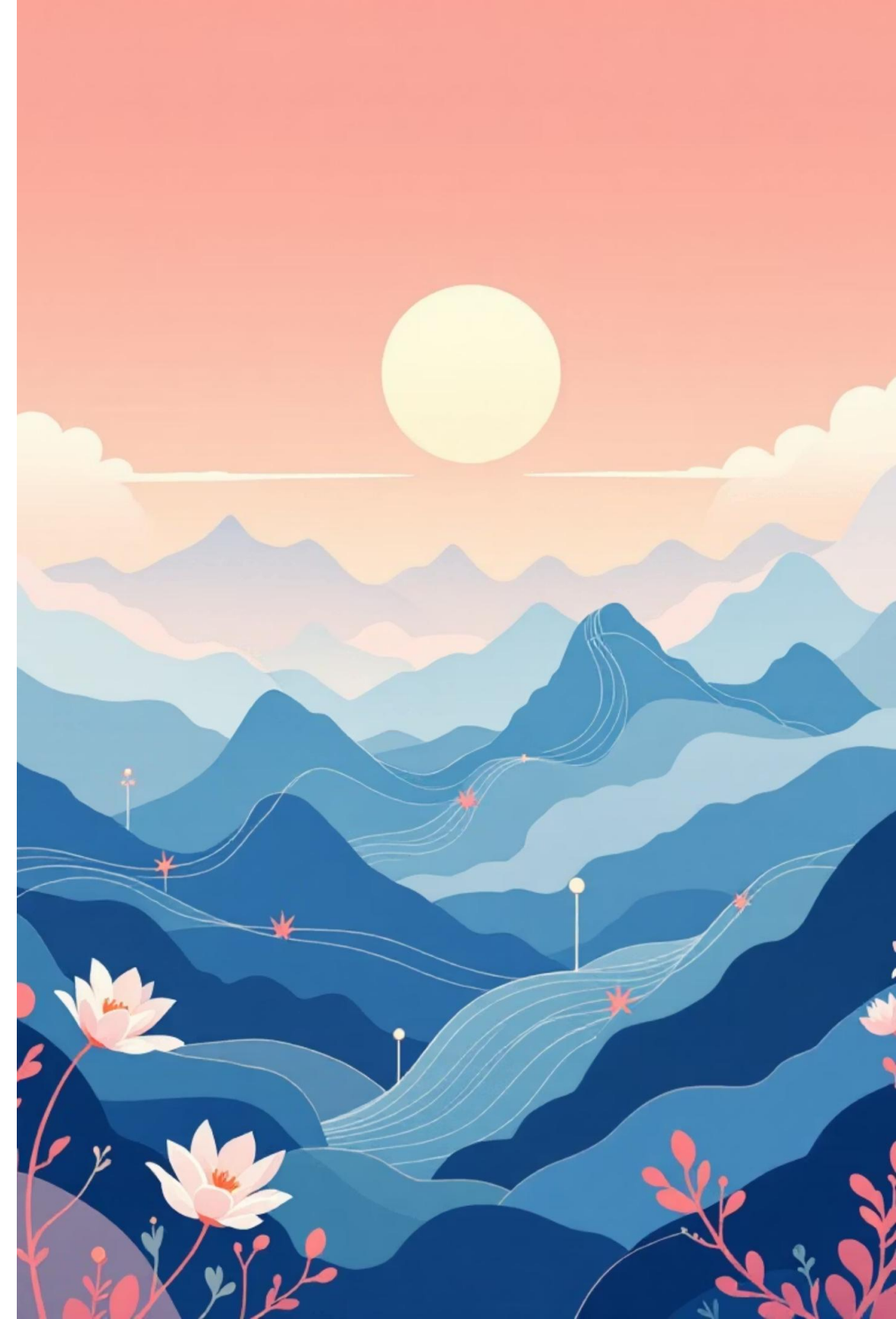
Differences between predictions and actual values reveal contextual anomalies.

4

Apply MAD Threshold

Median Absolute Deviation identifies significant deviations requiring attention.

This approach captures **contextual anomalies** — values that are unusual for a specific person at a specific time, even if they fall within general normal ranges.



Clustering-Based Anomaly Detection

Discovering Collective Behavioral Anomalies

What it Detects

- Identifies **collective / behavioral anomalies**
- Focuses on **unusual combinations of health metrics**, not single values

Why Clustering is Needed

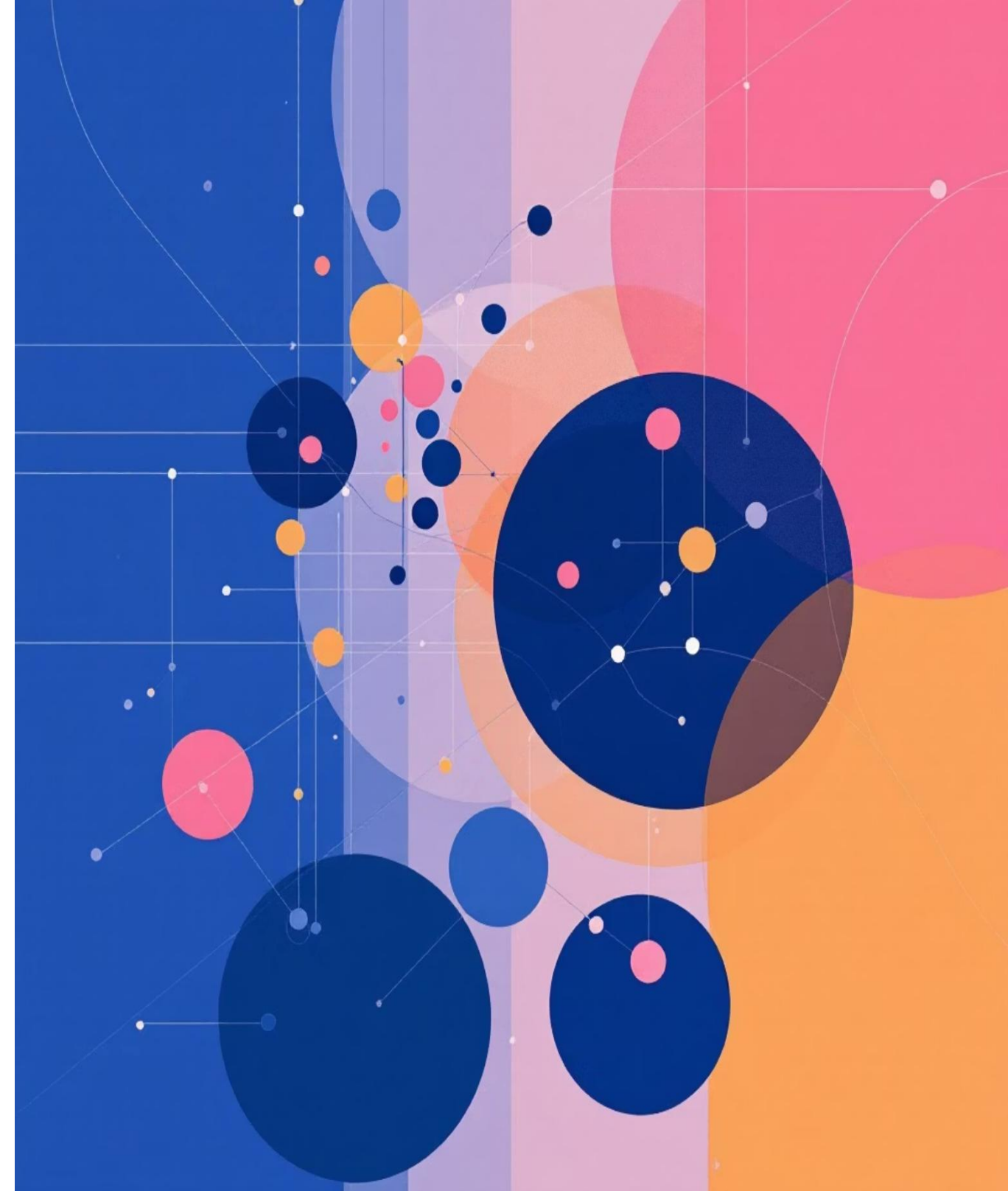
- Some anomalies are not extreme individually
- Abnormality appears only when **multiple features are considered together**

Method Used: DBSCAN

- Groups similar health behavior into clusters
- Automatically detects dense regions (normal behavior)
- Data points not belonging to any cluster are labeled as **noise**
- **Noise points (label = -1)** are treated as anomalies

Example

1. High heart rate + high steps → normal workout
2. High heart rate + zero steps → **anomalous behavior**
3. Low heart rate + very high steps → **behavioral inconsistency**



Interactive Anomaly Visualization

Why Visualization is Important

- Anomaly detection results are difficult to interpret from raw data tables
- Visualization converts numerical outputs into **clear, human-understandable insights**
- Helps identify **when, how often, and how severe** anomalies occur

Types of Visual Insights Provided

1. **Time-series views:**
Show normal heart rate patterns over time and highlight sudden deviations.
2. **Anomaly markers:**
Clearly distinguish abnormal points from normal data using visual markers.
3. **Pattern comparison:**
Compare expected (model-predicted) behavior with actual behavior to identify anomalies.
4. **Behavioral insights:**
Reveal unusual combinations of health metrics using clustering visualizations.



Milestone 4: Dashboard for Insights

Objective:

To present detected health anomalies through an interactive dashboard for easy interpretation and reporting

Implementation:

- Developed a **Streamlit-based web interface**
- Integrated anomaly detection outputs from previous milestones
- Enabled users to upload fitness data (**CSV / JSON**) and trigger analysis

Visualization & Reporting:

- Displayed time-series charts with **clearly highlighted anomalies**
- Used **Matplotlib and Plotly** for interactive visualization
- Provided option to **export analysis results** in CSV / PDF format

Outcomes:

- Built an **interactive Streamlit dashboard** for health data analysis
- Successfully visualized detected anomalies in heart rate, activity, and sleep data
- Enabled users to **upload fitness data and run anomaly detection** easily
- Provided **clear, interpretable time-series visualizations** with highlighted anomalies
- Allowed exporting of results in **CSV / PDF format** for reporting and further analysis

OUTPUTS:

```
... === Dataset Info ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2016 entries, 0 to 2015
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   timestamp       2016 non-null   object
 1   heart_rate_bpm  2016 non-null   int64
 2   sleep_hours     2016 non-null   float64
 3   step_count      2016 non-null   int64
 4   calories_burned 2016 non-null   float64
 5   activity_type   2016 non-null   object
dtypes: float64(2), int64(2), object(2)
memory usage: 94.6+ KB
None

=== First 5 rows ===
      timestamp  heart_rate_bpm  sleep_hours  step_count \
0  2025-12-10 00:00:00         51         0.08         7
1  2025-12-10 00:05:00         50         0.17         0
2  2025-12-10 00:10:00         59         0.25         7
3  2025-12-10 00:15:00         56         0.33         4
4  2025-12-10 00:20:00         56         0.42         5

      calories_burned activity_type
0          0.79      Sleeping
1          0.50      Sleeping
2          0.87      Sleeping
3          0.72      Sleeping
4          0.76      Sleeping

=== Column names ===
Index(['timestamp', 'heart_rate_bpm', 'sleep_hours', 'step_count',
      'calories_burned', 'activity_type'],
      dtype='object')

Number of duplicate rows: 0
```

```
... Data Generation:
SAMPLE DATASET FOR THIS PROJECT
      timestamp  heart_rate_bpm  sleep_hours  step_count \
0  2025-12-10 00:00:00         51         0.08         7
1  2025-12-10 00:05:00         50         0.17         0
2  2025-12-10 00:10:00         59         0.25         7
3  2025-12-10 00:15:00         56         0.33         4
4  2025-12-10 00:20:00         56         0.42         5

      calories_burned activity_type
0          0.79      Sleeping
1          0.50      Sleeping
2          0.87      Sleeping
3          0.72      Sleeping
4          0.76      Sleeping
```

Total records: 2016
Duplicate rows: 674

Numeric column statistics:

	heart_rate_bpm	sleep_hours	step_count	calories_burned
count	2016.000000	2016.000000	2016.000000	2016.000000
mean	78.291171	0.760417	34.625496	2.167917
std	24.178391	1.576654	39.740700	1.819007
min	45.000000	0.000000	0.000000	0.480000
25%	60.000000	0.000000	4.000000	0.760000
50%	69.000000	0.000000	9.000000	0.970000
75%	87.000000	0.020000	57.000000	3.142500
max	141.000000	6.000000	149.000000	7.260000

Activity type counts:

activity_type

Walking 588

Resting 588

Sleeping 504

Running 168

Workout 168

Name: count, dtype: int64

No missing values left – Data quality looks good!

Resampled data:

	heart_rate_bpm	sleep_hours	step_count	calories_burned
timestamp				
2025-12-10 00:00:00	54.833333	0.541667	5.250000	0.758333
2025-12-10 01:00:00	56.166667	1.541667	3.916667	0.718333
2025-12-10 02:00:00	54.083333	2.541667	4.000000	0.700833
2025-12-10 03:00:00	55.666667	3.541667	3.833333	0.710000
2025-12-10 04:00:00	54.583333	4.541667	4.250000	0.715833

activity_type

timestamp

2025-12-10 00:00:00 Sleeping

2025-12-10 01:00:00 Sleeping

2025-12-10 02:00:00 Sleeping

2025-12-10 03:00:00 Sleeping

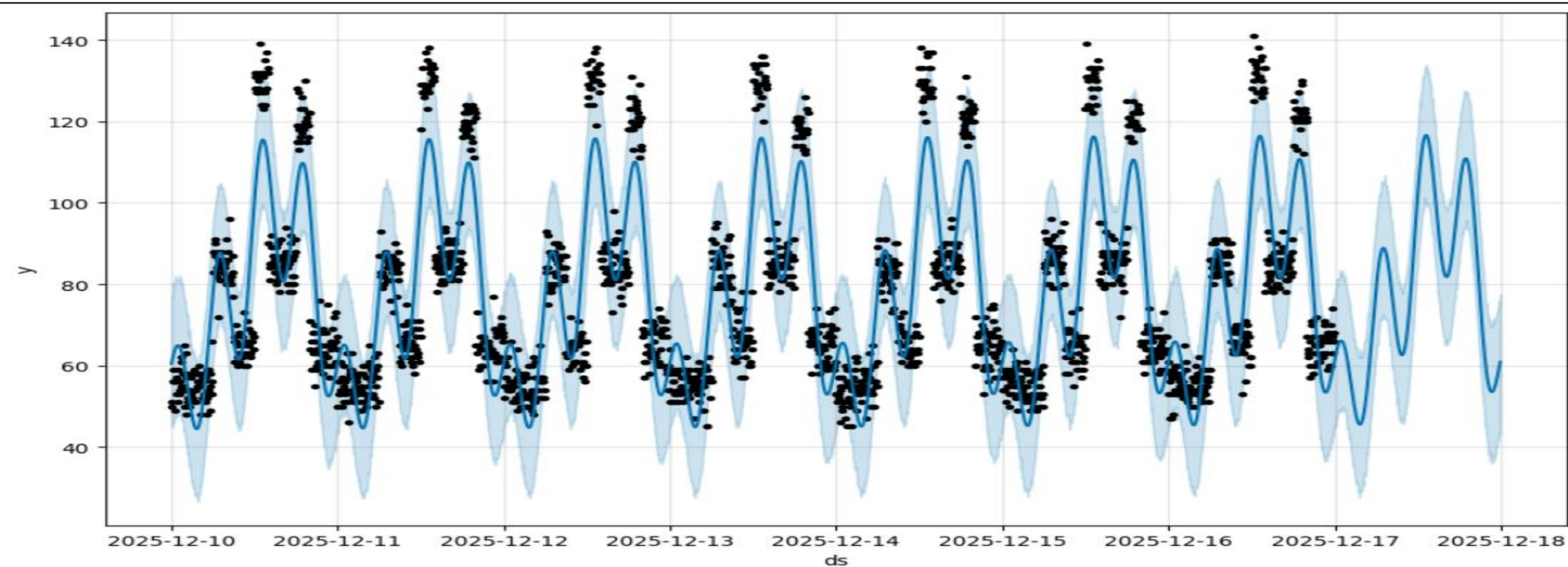
2025-12-10 04:00:00 Sleeping

Heart Rate Forecasting using Facebook Prophet

This graph shows heart rate data over time modeled using Facebook Prophet.

- **Black dots** represent actual heart rate values recorded by the fitness device.
- The **blue line** shows the predicted heart rate trend capturing daily seasonality.
- The **shaded blue region** represents confidence intervals around the prediction.

It helps identify **normal heart rate patterns**, and points deviating significantly from the trend can later be treated as **anomalies**.

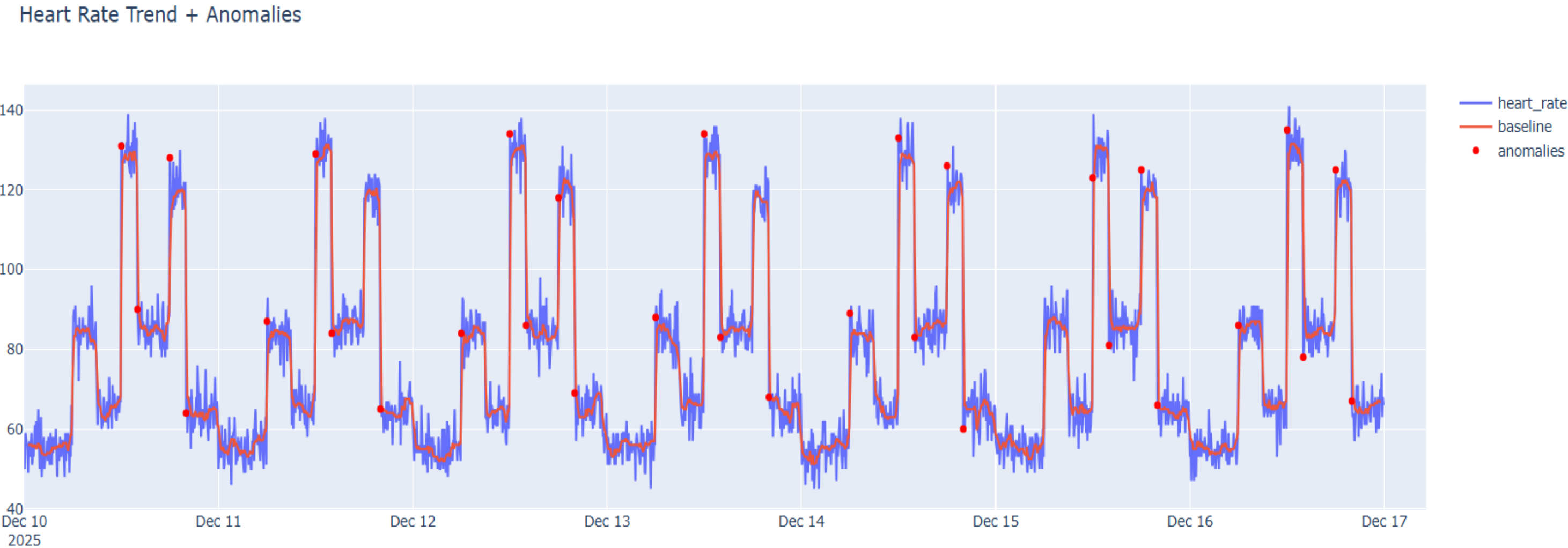


Heart Rate Trend with Detected Anomalies (Rolling Baseline Method)

This graph shows **heart rate behavior over time along with detected anomalies**.

- The **blue line** represents the actual heart rate values.
- The **orange line** shows the baseline trend (rolling median) representing normal behavior.
- The **red dots** indicate detected anomalies where the heart rate deviates significantly from the baseline.

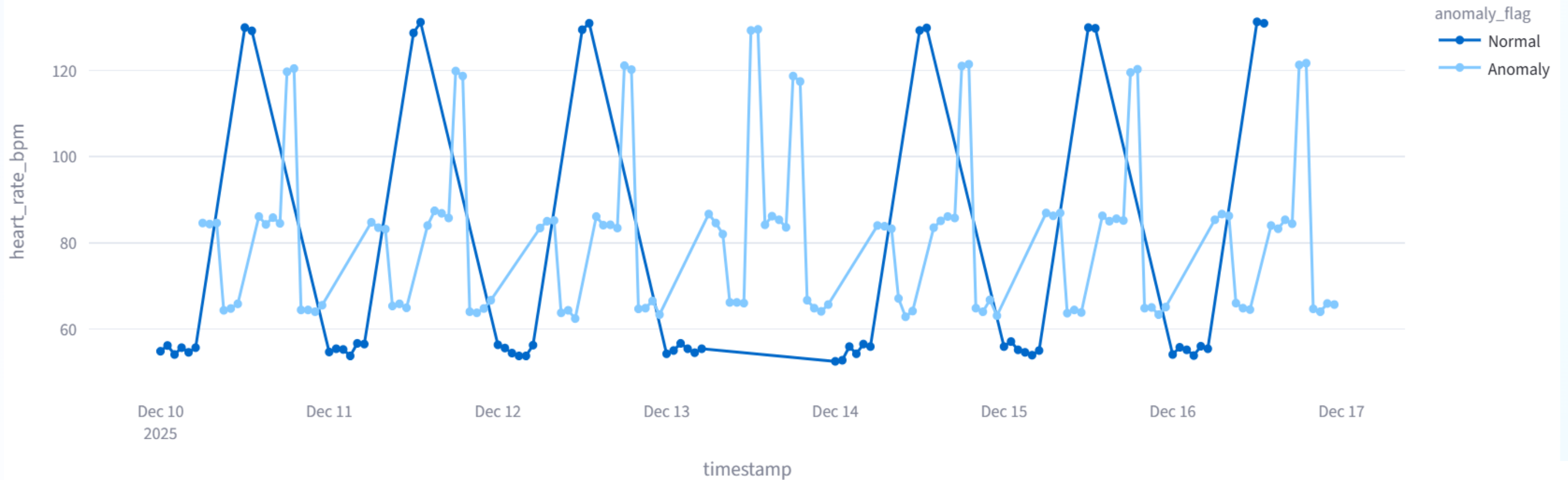
This visualization helps clearly identify **abnormal heart rate events** compared to normal daily patterns.



Heart Rate Timeline with Rule-Based Anomalies

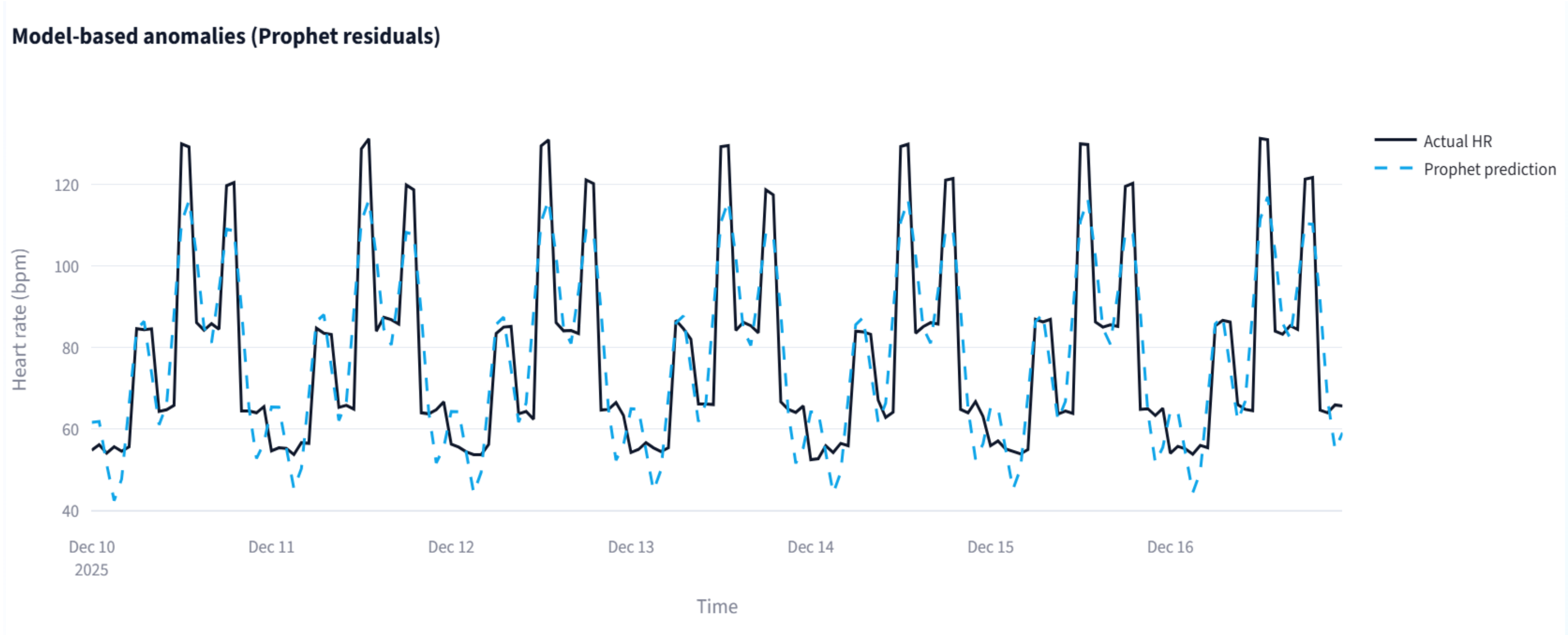
- The blue line shows **normal heart rate pattern over time**
- The highlighted points show **rule-based anomalies**
- These anomalies occur when heart rate crosses fixed thresholds

Heart Rate with Rule-based Anomalies



Model-Based Anomalies (Prophet Residuals)

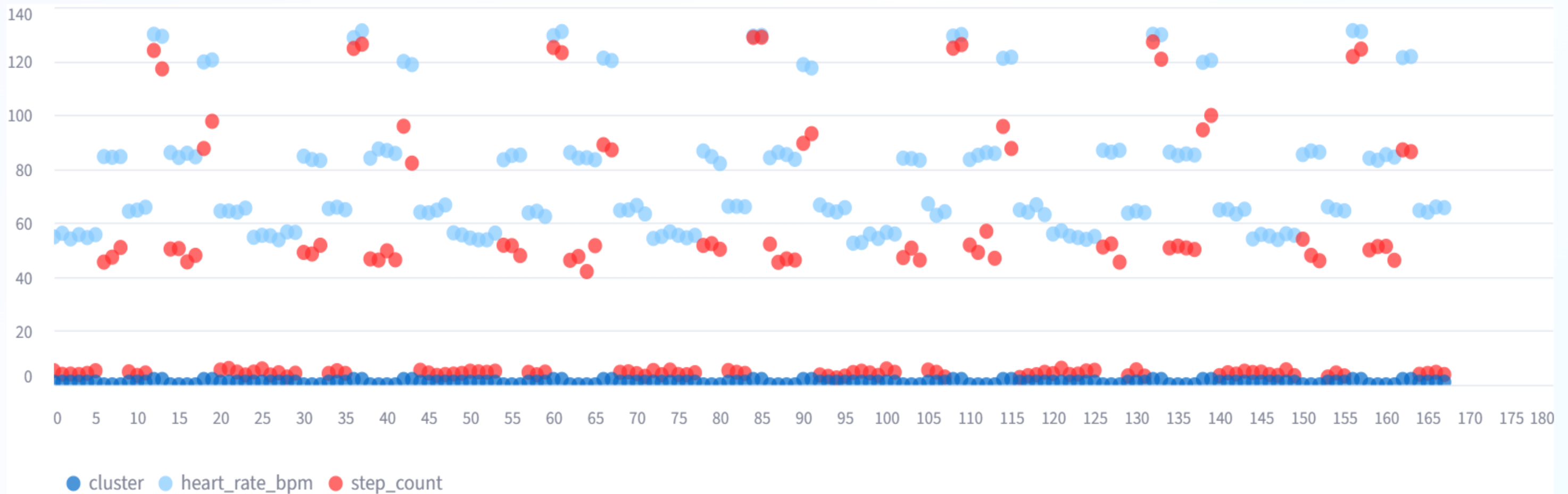
- Black line shows **actual heart rate**
- Dashed line shows **Prophet predicted heart rate**
- Large gaps between the two indicate **model-based anomalies**



K-Means Clustering (Heart Rate vs Steps)

- Data points are grouped into clusters based on **heart rate and step count**
- Each cluster represents a **common behavior pattern**
- Shows general activity groups like low, medium, and high activity

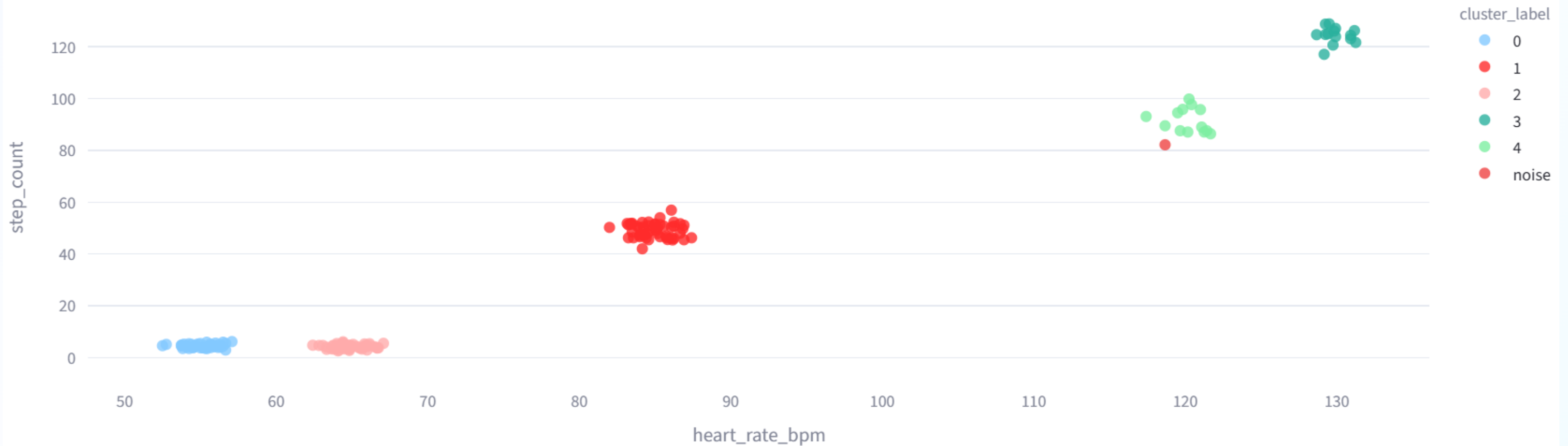
K-Means clustering (HR vs steps)



DBSCAN Clusters (Red = Noise)

- Normal behavior forms **dense clusters**
- Red points represent **noise points**
- Noise points are treated as **cluster-based anomalies**

DBSCAN clusters (red = noise)



Upload fitness CSV

Drag and drop file here

Limit 200MB per file • CSV

Browse files

Forecast hours ahead (Prophet)

24

Select view

- ☒ Overview
- ☐ Milestone 1 – Preprocessing
- ☐ Milestone 2 – Features & Modeling
- ☐ Milestone 3 – Anomalies & Clusters
- ☐ Milestone 4 – Insights & PDF
- ☐ Show full raw data

FitPulse – Health Anomaly Detection from Fitness

Devices

End-to-end time-series pipeline: generate or ingest fitness data, clean it, extract features with TSFresh, model normal behaviour using Prophet, and detect anomalies in heart rate, steps, sleep, and calories.

▼ Project milestones & tech stack

Milestone 1 – Data Collection & Preprocessing

- Clean and resample heart rate, steps, sleep, calories to hourly time-series.

Milestone 2 – Feature Extraction & Modeling

- Use TSFresh to compute statistical features per day.
- Forecast heart rate using Prophet.

Milestone 3 – Anomaly Detection & Clustering

- Rule-based and residual anomalies.
- K-Means and DBSCAN clustering to group behavioural patterns.

Milestone 4 – Dashboard Insights & Report

- Summarise anomalies and export a PDF report.

FitPulse Controls

Upload fitness CSV



Sample_data.csv

84.9KB

×

Forecast hours ahead (Prophet)

24

Select view

● Overview

- Milestone 1 – Preprocessing

● Milestone 2 – Features & Modeling

- Milestone 3 – Anomalies & Clusters

- Milestone 4 – Insights & PDF

☐ Show full raw data

Deploy

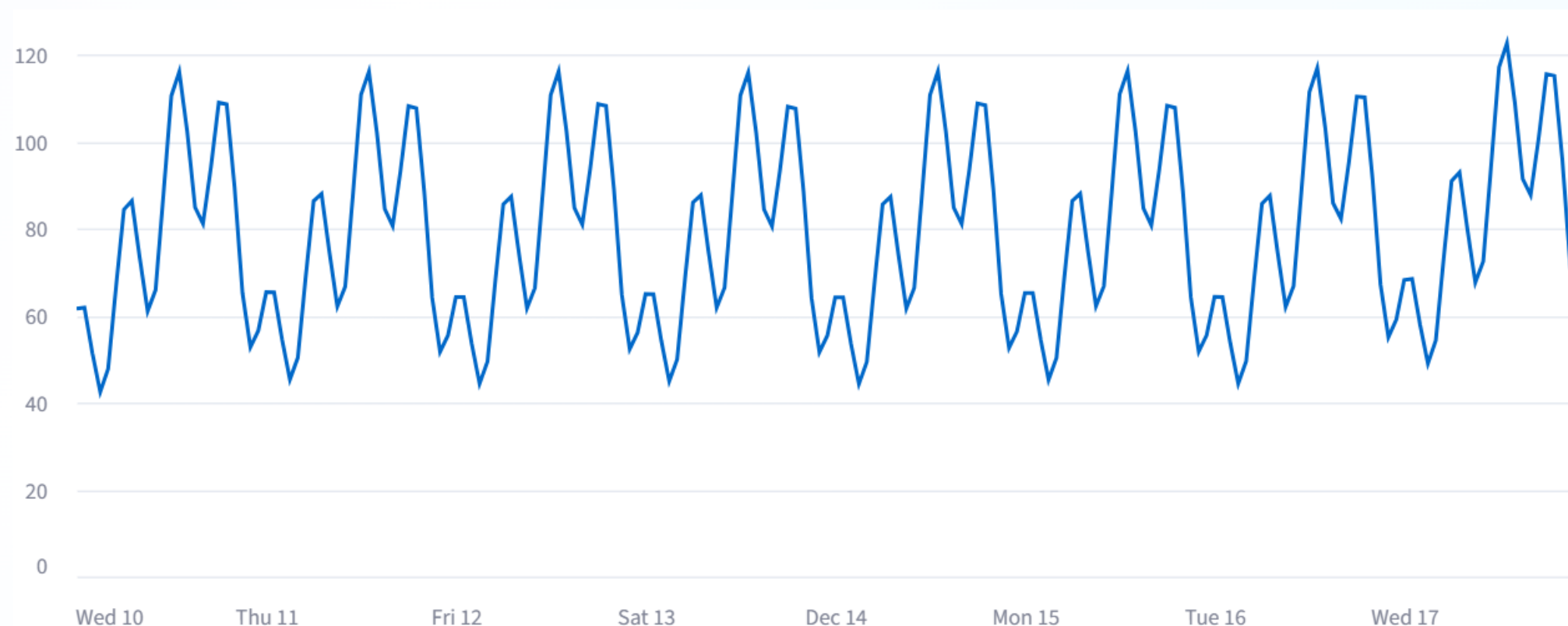


Milestone 2 – Feature Extraction & Modeling

TSFresh feature matrix and Prophet forecast.

Enable **Show TSFresh features** in the sidebar to inspect feature matrix.

Prophet forecast for heart rate



Conclusion & Future Directions

FitPulse successfully demonstrates an end-to-end pipeline for automated health anomaly detection from fitness device data. By combining time-series forecasting, clustering, and rule-based approaches, the system identifies irregular patterns that may indicate health concerns.

Key achievement: Early detection of health anomalies can empower users and healthcare providers to take preventive action before issues escalate, transforming passive data collection into active health monitoring.



Real-Time Data Streaming

Integrate with fitness device APIs for live anomaly detection and instant alerts



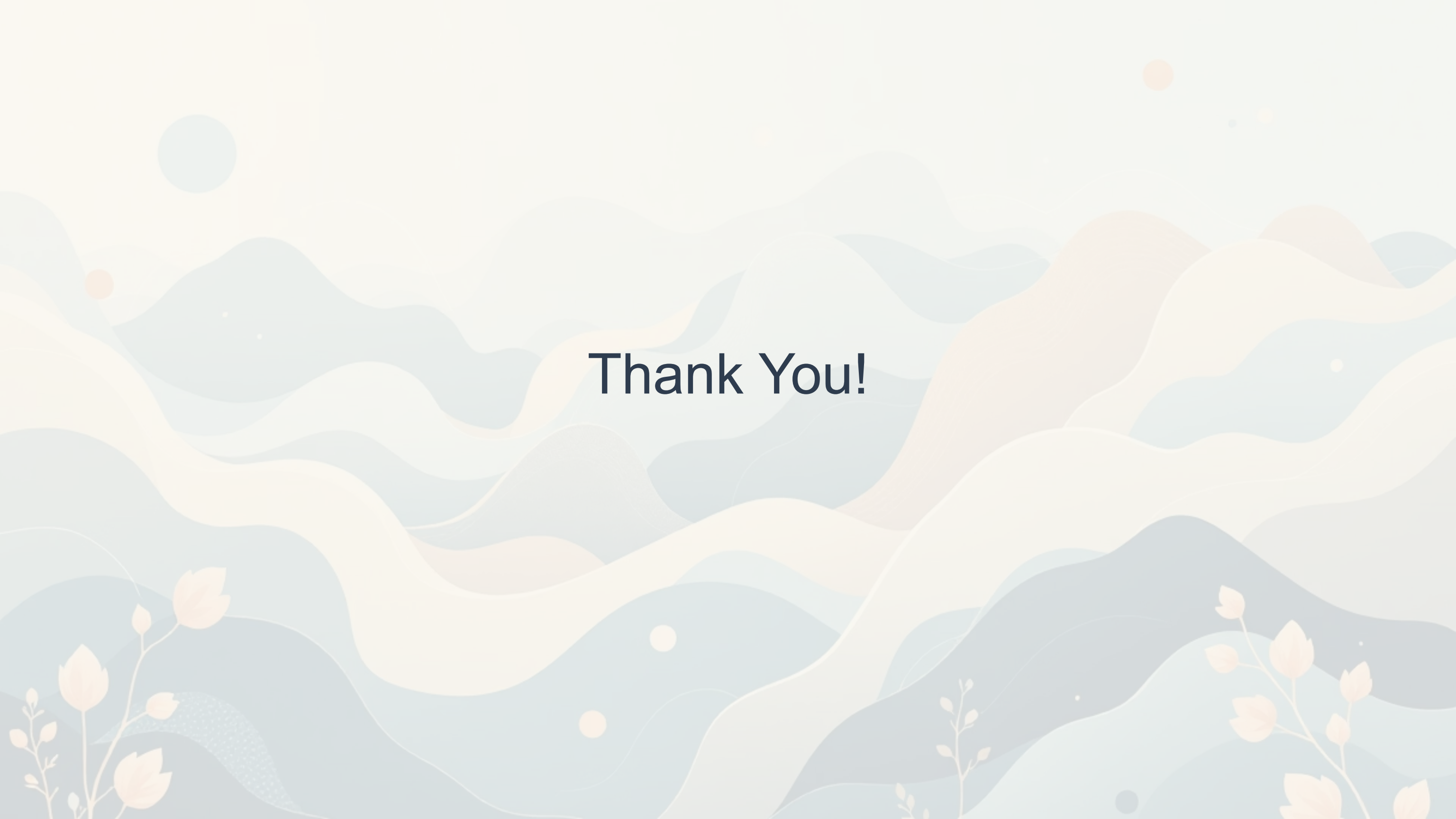
Deep Learning Models

Implement LSTM and Transformer networks for more sophisticated pattern recognition and forecasting



Personalized Thresholds

Adapt anomaly detection to individual baseline metrics, accounting for personal health profiles and fitness levels

The background features a series of overlapping, wavy, organic shapes in muted colors like sage green, dusty rose, and cream. Scattered throughout are small, solid-colored circles in these same tones. In the bottom left and right corners, there are delicate, stylized floral sprigs with small buds and leaves in a light cream color.

Thank You!